

Solving Problems Scientifically

A Comprehensive Technical Framework

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To TBD

*"An expert is a person who has made all the mistakes that can be made
in a very narrow field."* — Niels Bohr

Contents

Acknowledgements	ix
Preface	xi
Note to the Reader	xiii
List of Figures	xv
List of Tables	xvii
List of Equations	xix
1 The technical sandbox	1
1.1 Create figures with captions	1
1.2 Create bold, italics, and underline	1
1.3 Create URLs	1
1.4 Create lists	2
1.4.1 Create unordered lists	2
1.4.2 Create ordered lists	2
1.5 Create tables	2
1.5.1 Create simple tables	2
1.5.2 Create multi row tables	2
1.6 Create mathematics	3
1.6.1 Create inline mathematics	3
1.6.2 Create display style mathematics	3
1.7 Creating greek letters	4
1.7.1 Greek Unicode	4
1.7.2 Greek mathematics	5
1.7.3 Greek text	5
1.8 Software user interface	6
1.9 Inserting citations	6
1.10 Inserting glossary terms	6
1.11 Inserting index terms	6
1.12 Endnotes	7

2	Introduction	9
3	Scientific method	11
A	Python environment setup	13
A.1	Install Python for Arch Linux	13
A.1.1	Procedural steps	13
A.1.2	Install Python with pacman	13
A.1.3	Verify the version	13
A.1.4	Verify the environment	13
A.2	Install Python for Windows	14
A.2.1	Procedural steps	14
A.2.2	Verify the version	14
A.3	Install Python for macOS	14
A.3.1	Procedural steps	14
A.3.2	Verify the version	15
B	Appendix—dawgdad	17
B.1	Install dawgdad	17
B.2	Using dawgdad in Python	17
	Glossary	19
	Endnotes	33
	References	35
	Index	37
	About the Author	39

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Avtar Singh, who taught me persistence, drive, and the desire to learn anything.

And $\text{\LaTeX}$ and $\text{\LaTeX} 2_{\epsilon}$.

Preface

Why did I write this book? TBD

Note to the Reader

Work in Progress

Please note that this book is currently a **work in progress**. Content, code snippets, and structural frameworks are actively being developed, refined, and updated. Chapters may be added, reorganized, or revised as the material matures. There is a companion [YouTube channel](#). I do not have a timeline for this work. It is a labour of love.

Document Conventions

Links are shown in:

- [blue](#) for internal links to other parts within this book, such as chapters, sections, glossary, and page numbers.
- [red](#) for citation links to the reference section.
- [orange](#) for external links outside the book, such as a website, [YouTube channel](#), etc.

Software user interface

- [Caribbean mint](#) for selecting an item in a list.
- [neon pink](#) for text to enter.

Code and shell commands are shown in:

- Shell commands (zsh, bash) are shown inside a box outlined in black, with a note above it that identifies it as shell commands.
- Python commands are shown inside a box outlined in black, with a note above it that identifies it as code.

List of Figures

1.1 Anatomy of a matplotlib figure	1
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List of Tables

1.1	Visigothi cum Romanis	2
1.2	Comparison of capability indices	2
1.2	Comparison of capability indices	3
1.3	Greek Unicode	4
1.3	Greek Unicode	5
1.4	Greek mathematics	5
1.5	Greek text	6

List of Equations

1.1	Measured observation	3
1.2	Intraclass correlation coefficient	3
1.3	Average $\pm$ confidence interval	3
1.4	Distance to target	3
1.5	Standard deviation $\pm$ confidence interval	3
1.6	Distance to nearest specification	3
1.7	$\hat{P}_{pk}$ lower bound	4
1.8	$\hat{P}_{pk}$ upper bound	4
1.9	Matrix p format	4
1.10	Matrix b format	4

Chapter 1

The technical sandbox

This is a structural sandbox chapter designed to validate typography, math environments, glossary tracking, and custom list indexing. Use this file as a structural blueprint when spawning new chapters.

1.1 Create figures with captions

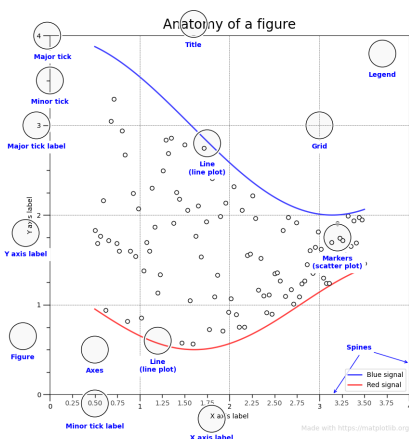


Figure 1.1: Anatomy of a matplotlib figure

1.2 Create bold, italics, and underline

Some of the **greatest** *discoveries* in science were made by ***accident***.

1.3 Create URLs

Hyperlinks within the document can be easily added. For example, [see Create figures with captions](#) uses `hyperref{}` to take you to a link based on a section or label. And `\ref{fig:labelname}` such as Figure [1.1](#) inserts the figure number as a link to that figure.

Hyperlinks to external URLs can be easily added. For example, see [Wikipedia L^AT_EX](#).

1.4 Create lists

1.4.1 Create unordered lists

- Illinois
 - Chicago
 - Naperville
- Kentucky
 - Lexington
 - Louisville

1.4.2 Create ordered lists

1. Cat
 - (a) Persian
 - (b) Siamese
2. Dog
 - (a) Aussiedoodle
 - (b) Newfoundland

1.5 Create tables

1.5.1 Create simple tables

Table 1.1: Visigothi cum Romanis

	I	II	III	IV	V	VI
Vandali	123	456	678	321	644	768
Visigothorum	021	229	678	123	456	678

1.5.2 Create multi row tables

Table 1.2: Comparison of capability indices

	Process requirement	Sampling	Comments
C_{pk}	- Stable and predictable variation - Normal distribution - Need at least one specification limit	- Rational - Representative of process - Ensure same process conditions - No identifiable condition changes	- Possible cost to increasing C_{pk} - Wide confidence interval
P_{pk}	- Stable and predictable variation - Normal distribution - Need at least one specification limit	- Random - Sufficiently large, representative - Ensure same process conditions - No identifiable condition changes	- Possible cost to increasing ppk - Wide confidence interval - Easiest formulae to explain
C_{pm}	- Stable and predictable variation - Normal distribution	- Same as C_{pk} or P_{pk}	- C_{pm} penalizes for being off-target - Wide confidence interval

Table 1.2: Comparison of capability indices

Process requirement	Sampling	Comments
- Need at least one specification limit - Need specification target		- Most difficult formulae to explain

1.6 Create mathematics

1.6.1 Create inline mathematics

Examples of inline mathematics $\frac{a}{b}$, followed by text and another inline equation $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and more text $\sum_{k=0}^{\infty} \frac{1}{2^k} = 2$ and finally much more text to cover many lines of a paragraph.

1.6.2 Create display style mathematics

Examples of mathematics on their own lines. All equations are flush-left to the page unless multilines, and thus aligned on a symbol with the longest equation flush-left. The equations numbers are flush-right.

$$\text{measured observation} = \text{true value} + \text{measurement system error} \quad (1.1)$$

$$\text{ICC} = \frac{\text{variance of the product values}}{\text{variance of the product observations}} \quad (1.2)$$

$$\bar{X} \pm \text{CI} = \frac{\sum_{i=1}^n X_i}{n} \pm t_{1-\alpha/2, n-1} \times \frac{s}{\sqrt{n}} \quad (1.3)$$

$$\text{distance to target} = \text{specification target} - \text{location statistic} = T - \bar{X} \quad (1.4)$$

$$s \pm \text{CI} = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}} \pm \left(\sqrt{\frac{(n-1)s^2}{\chi_{\alpha/2}^2}}, \sqrt{\frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}} \right) \quad (1.5)$$

$$\text{distance to nearest spec} = \frac{\text{nearest spec} - \text{location statistic}}{\text{dispersion statistic}} = \frac{\min(\bar{X} - \text{LSL}, \text{USL} - \bar{X})}{s} \quad (1.6)$$

The confidence intervals are:

$$\text{Lower bound} = \hat{P}_{pk} - z_{1-\alpha/2} \sqrt{\frac{1}{\left(\frac{\text{toler}}{2}\right)^2 n} + \frac{(\hat{P}_{pk})^2}{2\nu}} \quad (1.7)$$

$$\text{Upper bound} = \hat{P}_{pk} + z_{1-\alpha/2} \sqrt{\frac{1}{\left(\frac{\text{toler}}{2}\right)^2 n} + \frac{(\hat{P}_{pk})^2}{2\nu}} \quad (1.8)$$

where:

n = the number of observations in the sample

ν = the degrees of freedom = $n - 1$

$\chi^2_{\alpha,\nu}$ = the α percentile of the chi-square distribution with ν degrees of freedom

toler = multiplier of the standard deviation tolerance (Minitab® uses 6 as the default value)

$z_{1-\alpha/2}$ = the $1 - \alpha/2$ percentile of the standard normal distribution

$$A = \begin{pmatrix} 1 & 0 & 4 & -1 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (1.9)$$

$$A = \begin{bmatrix} 1 & 0 & 4 & -1 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.10)$$

1.7 Creating greek letters

1.7.1 Greek Unicode

Greek letters can be inserted in mathematics and text using Unicode characters. In Neovim, the sequence is 'ControlKey + v uhhhh' where 'hhhh' is the Unicode.

Table 1.3: Greek Unicode

letter	code	miniscule	code	majuscule
alpha	03b1	α	0391	A
beta	03b2	β	0392	B
gamma	03b3	γ	0393	Γ
delta	03b4	δ	0394	Δ
epsilon	03b5	ε	0395	E
zeta	03b6	ζ	0396	Z
eta	03b7	η	0397	H

Table 1.3: Greek Unicode

letter	code	miniscule	code	majuscule
theta	03b8	θ	0398	Θ
iota	03b9	ι	0399	Ι
kappa	03ba	κ	039a	Κ
lambda	03bb	λ	039b	Λ
mu	03bc	μ	039c	Μ
nu	03bd	ν	039d	Ν
xi	03be	ξ	039e	Ξ
omicron		ο		Ο
pi	03c0	π	03a0	Π
rho	03c1	ρ	03a1	Ρ
sigma	03c3	σ	03a3	Σ
tau	03c4	τ	03a4	Τ
upsilon	03c5	υ	03a5	Υ
phi	03c6	φ	03a6	Φ
chi	03c7	χ	03a7	Χ
psi	03c8	ψ	03a8	Ψ
omega	03c9	ω	03a9	Ω

1.7.2 Greek mathematics

Greek letters can be inserted in mathematics. These are naturally italicized.

Table 1.4: Greek mathematics

letter	code	miniscule	code	majuscule
alpha	<code>\alpha</code>	α	A	A
beta	<code>\beta</code>	β	B	B
gamma	<code>\gamma</code>	γ	<code>\Gamma</code>	Γ
delta	<code>\delta</code>	δ	<code>\Delta</code>	Δ
epsilon	<code>\epsilon</code>	ϵ	E	E
zeta	<code>\zeta</code>	ζ	Z	Z
eta	<code>\eta</code>	η	H	H
theta	<code>\theta</code>	θ	<code>\Theta</code>	Θ
iota	<code>\iota</code>	ι	I	I
kappa	<code>\kappa</code>	κ	K	K
lambda	<code>\lambda</code>	λ	<code>\Lambda</code>	Λ
mu	<code>\mu</code>	μ	M	M
nu	<code>\nu</code>	ν	N	N
xi	<code>\xi</code>	ξ	<code>\Xi</code>	Ξ
omicron	<code>o</code>	o	O	O
pi	<code>\pi</code>	π	<code>\Pi</code>	Π
rho	<code>\rho</code>	ρ	P	P
sigma	<code>\sigma</code>	σ	<code>\Sigma</code>	Σ
tau	<code>\tau</code>	τ	T	T
upsilon	<code>\upsilon</code>	υ	<code>\Upsilon</code>	Υ
phi	<code>\phi</code>	ϕ	<code>\Phi</code>	Φ
chi	<code>\chi</code>	χ	X	X
psi	<code>\psi</code>	ψ	<code>\Psi</code>	Ψ
omega	<code>\omega</code>	ω	<code>\Omega</code>	Ω

1.7.3 Greek text

Greek letters can be inserted as text.

Table 1.5: Greek text

letter	code	miniscule	code	majuscule
alpha	<code>\textalpha</code>	α	<code>\textAlpha</code>	A
beta	<code>\textbeta</code>	β	<code>\textBeta</code>	B
gamma	<code>\textgamma</code>	γ	<code>\textGamma</code>	Γ
delta	<code>\textdelta</code>	δ	<code>\textDelta</code>	Δ
epsilon	<code>\textepsilon</code>	ε	<code>\textEpsilon</code>	Ε
zeta	<code>\textzeta</code>	ζ	<code>\textZeta</code>	Z
eta	<code>\texteta</code>	η	<code>\textEta</code>	H
theta	<code>\texttheta</code>	θ	<code>\textTheta</code>	Θ
iota	<code>\textiota</code>	ι	<code>\textIota</code>	I
kappa	<code>\textkappa</code>	κ	<code>\textKappa</code>	K
lambda	<code>\textlambda</code>	λ	<code>\textLambda</code>	Λ
mu	<code>\textmu</code>	μ	<code>\textMu</code>	M
nu	<code>\textnu</code>	ν	<code>\textNu</code>	N
xi	<code>\textxi</code>	ξ	<code>\textXi</code>	Ξ
omicron	<code>\textomicron</code>	ο	<code>\textOmicron</code>	Ο
pi	<code>\textpi</code>	π	<code>\textPi</code>	Π
rho	<code>\textrho</code>	ρ	<code>\textRho</code>	Ρ
sigma	<code>\textsigma</code>	σ	<code>\textSigma</code>	Σ
tau	<code>\texttau</code>	τ	<code>\textTau</code>	T
upsilon	<code>\textupsilon</code>	υ	<code>\textUpsilon</code>	Υ
phi	<code>\textphi</code>	φ	<code>\textPhi</code>	Φ
chi	<code>\textchi</code>	χ	<code>\textChi</code>	Χ
psi	<code>\textpsi</code>	ψ	<code>\textPsi</code>	Ψ
omega	<code>\textomega</code>	ω	<code>\textOmega</code>	Ω

1.8 Software user interface

Here is for [selecting an item](#) in a list. Here is for [user input](#).

1.9 Inserting citations

We also anchor technical references using our external database system (Wheeler and Lyday [1989](#)). Here is a single citation without prenote and postnote. There is a definitive book on MSA (Wheeler [2006](#)). Here is a single citation with prenote only. There is a definitive book on SPC (Wheeler [2006](#)). Here is a single citation with postnote only. C_{pm} is a complex subject (Chan, Cheng, and Spiring [1988](#), pp. 1–10).

1.10 Inserting glossary terms

For glossary terms use [endogenous variable](#), [endogenous variables](#), [exogenous variable](#), [exogenous variables](#), [machine learning](#), and [seborrhic keratosis](#).

1.11 Inserting index terms

The development of advanced automation frameworks requires a tight integration between custom infrastructure and robust software engineering. When compiling highly optimized modules, developers must pay close attention to memory constraints.

1.12 Endnotes

The initial deployment of machine learning architectures required massive mainframe hardware systems.¹ This structural limitation heavily restricted real-time calculations.

Chapter 2

Introduction

Solving problems scientifically involves using the **scientific method**, a structured, data-driven framework designed to eliminate guesswork and replace it with empirical evidence. [ADD MORE](#)

This is a test for indexing. This is a sentence discussing automated engineering solutions and high-performance data processing software.

Chapter 3

Scientific method

The scientific method is doing something in a systematic and logical manner. A claim is verified through rigorous trails, not anecdotal evidence.

From one source:

- observation
- hypothesis
- experimentation
- conclusion

Other thoughts:

- what problem are you trying solve?
- establish methods of monitoring and control

What does Wikipedia say?

What does the Cambridge Dictionary say?

Appendix A

Python environment setup

A.1 Install Python for Arch Linux

On Arch Linux, Python is treated as a core system component. Do not install manual binaries from the website; instead, use the native [package manager](#), [pacman](#), to ensure rolling-release dependency tracking remains unified.

A.1.1 Procedural steps

1. Synchronize the package databases and upgrade the system infrastructure.
2. Explicitly install the core Python package.

A.1.2 Install Python with pacman

Identify as shell commands:

```
$ Shell: pacman
```

```
sudo pacman -Syu python
```

A.1.3 Verify the version

Validate the native binary deployment:

```
$ Shell: Python version
```

```
python --version
```

To exit the interactive environment, execute **exit()**.

A.1.4 Verify the environment

Regardless of your underlying host operating system, a successful deployment should allow entry into the interactive Python Read-Eval-Print Loop (REPL).

```
$ Shell: Run Python
```

```
python
```

Expected terminal output state:

\$ Shell: Python version

```
Python 3.14.6 (main, Jun 15 2026, 11:36:54) [GCC 16.1.1 20260430] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>>
```

To exit the interactive environment, execute **exit()**.

A.2 Install Python for Windows

To establish a clean Python environment on Microsoft Windows, utilize the official installer provided by the Python Software Foundation.

A.2.1 Procedural steps

1. Navigate to the official download architecture at python.org.
2. Download the latest stable **Windows installer (64-bit)**.
3. Launch the executable installer.
4. **Critical Step:** At the bottom of the installation wizard window, check the box labeled **"Add python.exe to PATH"**. Failure to do this will prevent execution from standard terminals.
5. Select **"Install Now"**.
6. Once complete, verify the environmental path configuration by opening **PowerShell** or the Command Prompt and executing the verification command.

A.2.2 Verify the version

Identify as shell commands:

\$ Shell: Python version

```
python --version
```

To exit the interactive environment, execute **exit()**.

A.3 Install Python for macOS

While macOS includes a system stub for Python, development workflows require a dedicated, unencumbered framework binaries from the official source.

A.3.1 Procedural steps

1. Navigate to python.org.
2. Download the **macOS universal installer** package (.pkg).
3. Run the installer package and follow the standard system prompts.
4. The installer automatically handles path updates for standard shells.

A.3.2 Verify the version

Open your terminal (Zsh) and execute:

```
$ Shell: Python version
```

```
python3 --version
```


Appendix B

Appendix—dawgdad

B.1 Install dawgdad

dawgdad (Pilon 2026) is a **Python** package for statistical, graphical, and predictive analysis.

dawgdad requires Python. Install a **Python** distribution appropriate to your operating system (**Python environment setup**).

Install dawgdad following the instructions at <https://github.com/gillespilon/dawgdad>. dawgdad is compatible with the **latest Python distribution** from **Python.org**. Instructions on how to use dawgdad are provided in relevant sections of this document. Full documentation is available at <https://dawgdad.readthedocs.io/en/latest/>.

Run the following command in your shell:

\$ Shell: Installing dawgdad

```
pip install dawgdad
```

If you prefer a different distribution, such as **Anaconda**, following the instruction as appropriate.

B.2 Using dawgdad in Python

Here is a quick example of how to import and initialize the module:

▷ **Code: Initializing the data framework**

```
1 import dawgdad as dd
2
3 # Load and analyze your dataset
4 data = dd.load_analytics("performance_log.csv")
5 print(data.summary())
```


Glossary

Symbols

1d array One-dimensional array. A NumPy array whose *.shape* has length 1. A vector.

2d array Two-dimensional array. A NumPy array whose *.shape* has length 2. Often represents a matrix.

A

abbreviation It is a shortened form of a word.

acronym It is a pronounceable word composed of a series of initial letters or parts of words.

ad hominem Instead of dealing with an idea, one tries to denigrate the messenger.

additive An additive model of *exogenous variables* implies that the observed effects are due to main effects only and no interactions (Box, W. G. Hunter, and J. S. Hunter 1978, p. 218).

agentic AI TBD *see also* *AI agent*

AI agent An AI agent is a software application that focuses on autonomous systems that can make decisions and perform tasks without human intervention. The independent systems automatically respond to conditions, to produce process results. An AI agent operates independently, making decisions through continuously learn and analysis of external data and complex data sets. Functioning agents can require various AI techniques, such as *natural language processing*, *machine learning*, and *computer vision*.

AI chatbot An AI chatbot is a software application designed to have textual or spoken conversations. It uses *natural language processing*, *machine learning* algorithms, and *generative artificial intelligence* systems that are capable of maintaining a conversation with use in *natural language* and simulating the way a human would behave as a conversation partner.

amortization Amortization spreads out the cost of an intangible assets over its useful life. Do not count 'goodwill amortization' because it is not an economic expense. It is a non cash charge to account for the loss of value of an intangible assets over time. 22

analysis of main effects The analysis of main effects (ANOME) chart that compares the *average* of each level to the overall average

analysis of mean ranges The analysis of mean ranges (ANOMR) chart compares the average range of each level to the overall range

application programming interface It is a way for two or more *computer programs* to communicate with each other. It is a type of software interface, offering a service to other pieces of software. 25, 31

Arch Linux Arch Linux is a *open-source software*, rolling release Linux distribution. (Arch Linux) 27

argument Arguments are the values assigned to *parameters* when passed into a Python function or method during a function call.

artificial general intelligence Artificial general intelligence (AGI) means machines will be endowed with intelligence as good as the smartest human in any field, but because of its capabilities to integrate learning across many fields, it will probably become better than any average doctor, lawyer or computer programmer.

artificial intelligence Artificial intelligence (AI) is about creating computer systems that can perform tasks that typically require human intelligence, such as understanding language, recognizing images, and making decisions. It is teaching computers to think and learn like humans. 19, 21, 23, 25, 28, 29, 31

artificial narrow intelligence Narrow artificial intelligence is limited to a single, narrowly defined task. Most modern *artificial intelligence* systems are of this type. It is also called weak artificial intelligence.

artificial neural network It is a network of artificial neurons or nodes. 21, 23-25, 31

artificial super intelligence Artificial super intelligence (ASI) is a computer brain that can exceed what any human can do in any field and then, with its *polymathic* ability, it could produce insights far beyond anything humans could do or even imagine. It might even invent its own language we do not understand.

average It is the sum of a set of values divided by the count of values. It is a type of **location statistic**.

$$\bar{Y} = \frac{\sum_{i=1}^n X_i}{n}$$

19, 20, 22, 24–26, 28, 29, 32

average absolute deviation The average absolute deviation (AAD) measure does not square the distance from the average, so it is less affected by extreme **observations** than are the **variance** and the **standard deviation**.

$$\text{AAD} = \frac{\sum_{i=1}^N (|Y_i - \bar{Y}|)}{N}$$

22, 26

average chart ($\bar{X}$) It is a **process behaviour chart** that plots the **average** of a subgroup of values taken in a rational manner. The chart also has an upper **control chart limit** line, a lower **control chart limit** line, and an **average** line. 20

average range It is the **average** of several **ranges** of sample subgroups. A subgroup is sampled, a **range** is calculated, and the **average** of the **ranges** is calculated. 32

B

Beautiful Soup It parses HTML and XML documents to create a tree-like representation of the document structure and extract **data**. (Richardson 2024)

between-operator error It is the variation in the **average** of measurements made by different operators measuring one characteristic on the same parts. It can be caused by differences in training, education, skill, and care in taking the measurements.

between-part error It is the variation of measurements between samples of parts, that is, the part-to-part variation. If a sample is carefully chosen, the error is shown on the **individuals chart** (X) or the **average chart** ($\bar{X}$).

between-subgroup error The subgroup averages are used to find an indirect estimate of the **standard deviation** of X .

bias and linearity This is the difference between the measurement result and its unknown, true values. It can be eliminated by calibration to a standard. Bias is an **average** over all reference values. Linearity or non-linearity is a difference in bias across all reference values.

biometric device A biometric device is an electronic device that uses biometric identifiers to identify and verify individuals.

biometric identifier A biometric identifier is a unique physical characteristic that can be used to identify an individual. Examples include fingerprint, retina or iris, voiceprint, hand geometry, face geometry, ear geometry, and heartbeat.

blocking It is groups of experimental runs in which randomisation is restricted within a group, for example, four runs in the morning and four runs in the afternoon. The relative homogeneity within a block and the identification and inclusion of the block as a factor in the **experiment** can be used to increase the accuracy of an **experiment**.

C

cappuccino Cappuccino is **espresso** coffee topped with frothed hot milk or cream and often flavored with cinnamon.

chi-square distribution It is the distribution of a sum of the squares of k independent standard normal random variables with k degrees of freedom.

coder A coder translates a problem into a language that the computer understands and creates a solution to that problem.

common cause Variation due to common cause is expected and natural variation. 21

computational linguistics Computational linguistics is an interdisciplinary field concerned with the computational modeling of **natural language**, as well as the study of appropriate computational approaches to linguistic questions.

computer A computer is a machine that executes a **computer program** to carry out sequences of operations to perform a set of tasks.

computer program A computer program is a sequence of instructions written in a **programming language** that is executed by a computer. 19, 20, 26–28, 30

computer vision Computer vision includes methods for acquiring, processing, analyzing, and understanding digital images, and extraction of high dimensional data from the real world in order to produce numerical or symbolic information, for example, in the form of decisions. 19, 29

conditional probability TBD 22, 23

confidence interval In frequentist **statistic**, it is a range of estimates for an unknown parameter, interval calculated from values in a sample.

confounding It is a column of an orthogonal array containing more than one effect which cannot be separated without further experimentation.

confusion matrix In the field of **machine learning**, it's a table that shows the performance of an algorithm, typically a supervised learning model. Each row of the matrix represents the instances in an actual class while each column represents the instances in a predicted class.

control chart It is a graphical plot used to determine if process variation is stable and predictable. A process that is in **statistical control** means that the variation is due to **common causes**. A process that is not in **statistical control**, that is, out of control, means that the variation is due to **special causes**.

control chart limit This is a limit of predictability for the **data** plotted on the relevant control chart. It is usually set at three **standard deviations** from a measure of **location**. 20, 24, 26, 29

conversational artificial intelligence Conversational artificial intelligence refers to any **artificial intelligence** system designed to interact with users through **natural language** conversation.

cybersecurity It is the protection of computer systems and networks from threats that may result in unauthorized information disclosure, theft of or damage to hardware, software, and **data**, and disruption or misdirection of the services they provide. 23, 27

D

data analyst This individual can quickly get a sense of what is worth pursuing with more care in an existing **data set**. Fast and humble, and sloppy code are acceptable traits of this role. This individual uses **statistic** and visualizations to explore the **data**, without coming to conclusions or decisions. 21

data engineer A data engineer prepares **data** for analytical and operational uses, that is, **extract, transform, and load**. This individual is capable of preparing **data** at scale from multiple sources and making it available to use in a production environment. 21

data science It is the science of extracting knowledge and insights from **data**. 21

data science architect The **data science** architect designs the collection, storage, and analyses processes, while considering time and cost-tradeoffs, and business requirements. This individual has expertise as a **data engineer**, **data analyst**, **machine learning engineer**, and **statistician**.

data scientist A data scientist applies the methodologies and tools of **data science**. This individual has expertise as a **data analyst**, **machine learning engineer**, and **statistician**.

data set A data set is a collection of **data**. In the case of tabular **data**, a data set corresponds to one or more database tables, where every column of a table represents a particular variable, and each row corresponds to a given record of the data set in question. The data set lists values for each of the variables, such as for example height and weight of an object, for each member of the data set. Data sets can also consist of a collection of documents or files. 21–23, 25, 26, 31, 32

dataleakage In **machine learning**, it is learning something from the **test data set** that one is not allowed to know.

datum A datum is an individual value in a collection of data, discrete or continuous values that convey information, describing the quantity, quality, fact, **statistic**, other basic units of meaning, or simply sequences of symbols that may be further interpreted formally. Data are usually organized into structures such as tables that provide additional context and meaning, and may themselves be used as data in larger structures. 20–23, 25–28, 30–32

dawgdad It is a Python package for statistical, graphical, and predictive analysis. To use dawgdad, install a Python distribution appropriate to your operating system. Then install dawgdad following the instructions at **dawgdad**.

decision maker It is someone who understands the art and science of data-driven decision-making. This individual is responsible for identifying decisions worth making with **data**, framing them (everything from designing metrics to calling the shots on statistical assumptions), and determining the required level of analytical rigour based on potential impact on the business.

deep learning It is a type of **machine learning** using an **artificial neural network** containing multiple **hidden layers**. 21, 25

deep model See **deep learning**.

deep neural network See **deep learning**.

deprecation A document has been superseded. It is recommended to not use the older document. It may not conform to current standard. It may contain errors or incorrect information. It may contain links to docu-

ments that no longer exist or their location is unknown. The content may be ambiguous or confusing to users.

depreciation It is a business expense. At some point a physical asset needs repairs or becomes obsolete. Money is spent before one gets revenue. A business should typically spend more than its depreciation to repair or acquire physical assets. Depreciation spreads out the cost of a tangible asset over its useful life. It is a non cash charge to account for the loss of value of a tangible assets over time. 22

descriptive analytics It seeks to get inspired with an existing data set, to summarize with statistic and visualizations, to look for patterns for possible further analysis using machine learning. 27

designed experiment It is a method to change factors simultaneously (inputs) using a structured pattern, in order to determine their effect on some response(s) (output(s)). 23, 24, 26, 29

digital human See virtual human

digital person See virtual human

digital transformation It is a process that aims to improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies.

discriminative models In statistical classification a discriminative approach computers a classifier such as logistic regression. A discriminative model is a model of the conditional probability $P(Y|X = x)$ of the target Y given an observation x . It can be used to discriminate the value of the target variable Y given an observation x .

dispersion A sample of values will not (usually) have all values identical. Dispersion is the variation between values. Large dispersion means the values are widely scattered. Small dispersion means the values are narrowly scattered. 22, 28, 32

dispersion statistic It is a statistic that estimates the spread or variability of the data. It is also called a measure of scale. Examples include the variance, the standard deviation, the range, the average absolute deviation, the median absolute deviation, and the interquartile range. 24

distance to nearest specification (sigma level) Is the difference between the location statistic and the closest specification limit, divided by the dispersion statistic (e.g. the standard deviation).

distance to target The difference between the specification target and the average is the difference you must adjust to aim the process correctly.

distribution package A distribution package is a version controlled archive file that contains Python packages, modules, and other resources files that are used to distribute a release. The archive file is what an end-user will download from the internet and install. A distribution package is a piece of software that you can install. Most of the time, this is synonymous with "project." When you type `pip install pkg`, or when you write `dependencies = ["pkg"]` in your `pyproject.toml`, `pkg` is the name of a distribution package. When you search or browse the PyPI, the most widely known centralized source for installing Python libraries and tools, what you see is a list of distribution packages.

docstring It is the embedded documentation for a Python module, class, function etc., usually in code as a string at the beginning of the object's definition.

dot plot It is a chart in which dots are used to depict the quantitative values (e.g. counts) associated with categorical variables.

E

ear geometry TBD

earnings TBD 22

EBITDA Earnings Before Interest Taxes Depreciation and Amortization. If the EBITDA is not stated, one can calculate EBIT from earnings (net income) + non operating expense + income taxes on the income statement, and DA from non cash charges on the cash flow statement. EBITDA is typically used by private equity companies to inflate their cash flow in order to obtain loans for business acquisitions.

effect It is the change in response caused by a change in the level of a factor (Hicks 1982, p. 114). 24

embedded operating system An embedded operating system is an operating system designed and optimized to improve the efficiency of managing the hardware resources and reduce response times specifically for the task the device was created for.

embedded system An embedded system is a specialized computer system; a combination of a computer processor, computer memory, and peripheral devices for input and output, that has a dedicated function within a larger mechanical or electronic system. It is embedded as part of a complete device often including electrical or electronic hardware and mechanical parts. Because an embedded system typically controls physical operations of the machine that it is embedded within, it often has real-time computing constraints. Embedded systems control many devices in common use.

empirical method It is any procedure for conducting an investigation that relies upon experimentation and systematic observation rather than theoretical speculation. 30

endogenous variable Endogenous variables are explained or predicted by a model. They are caused by factors within the system. Other names are y , dependent variable, regressand, outcome, response variable, and target ([machine learning](#)). In $y = mX + b$, y is the endogenous variable. [6](#), [25](#), [31](#)

endpoint detection and response Endpoint detection and response (EDR) is [cybersecurity](#) technology that continually monitors an endpoint (mobile phone, laptop, internet of things devices) to mitigate malicious cyber threats

espresso Espresso is coffee that is brewed by forcing hot water under high pressure through finely ground coffee beans. [20](#)

exogenous variable Exogenous variables are terms in a model. They are caused by factors outside the system. Other names are x , independent variable, regressor, design, explanatory variable, and feature ([machine learning](#)). In $y = mX + b$, x is the exogenous variable. In a [designed experiment](#), it is a factor that you can control, want to control, or is controlled at low cost. [6](#), [19](#), [23–25](#), [31](#)

expected mean squares They are the expected values of certain [statistic](#) arising in partitions of sums of squares in the analysis of variance. They can be used for ascertaining which statistic should appear in the denominator in an F test for testing a null hypothesis that a particular effect is absent.

experiment It is a series of [observations](#) conducted under controlled conditions to study a relationship with the purpose of drawing causal inferences about that relationship. An experiment involves the manipulation of an independent variable, the measurement of a dependent variable, and the exposure of various participants to one or more of the conditions being studied. Random selection of participants and their random assignment to conditions also are necessary in experiments. [20](#), [23](#), [27](#), [29](#), [30](#)

extended detection and response Extended detection and response (XDR) provides security incident detection and automated response capabilities for security infrastructure. It integrates threat intelligence and telemetry [data](#) from multiple sources with security analytics to provide conceptualization and correlation of security alerts.

extract, transform, and load At a high level extract, transform, and load (ETL) extracts, transforms, and loads [data](#). The specific tasks include: building [data](#) pipelines to gather information from difference source systems; integrate, consolidate, and clean data; structure the [data](#) for use in individual analytics applications. [21](#)

F

face geometry TBD

factor It is an independent variable that may be changed from treatment to treatment.

factor level It is the value that a factor in a [designed experiment](#) is set to.

factorial experiment It is a [designed experiment](#) that measures all possible combinations of factors. [23](#)

feature engineering It is the process of determining which [exogenous variables](#) might be useful in training a model, and then converting raw [data](#) into said [exogenous variable](#). It is useful to have domain knowledge.

fingerprint A fingerprint is the impression of a fingertip on any surface.

fixed effects model It is a statistical model in which the model parameters are non-random quantities. The mathematical model that describes the relationship between the response and treatment for the one-way ANOVA is given by $Y_{ij} = \mu + \tau_i + \epsilon_{ij}$, where Y_{ij} represents the j^{th} [observation](#) ($j = 1, 2, \dots, n_i$) on the i^{th} treatment ($i = 1, 2, \dots, k$ levels). μ is the common effect for the whole [experiment](#), τ_i represents the i^{th} treatment effect, and ϵ_{ij} represents the random error present in the j^{th} [observation](#) on the i^{th} treatment. The errors ϵ_{ij} are assumed to be normally and independently (NID) distributed, with mean zero and variance σ_e^2 . μ is always a fixed parameter, and $\tau_1, \tau_2, \dots, \tau_k$ are considered to be fixed parameters if the levels of the treatment are fixed and not a random sample from a population of possible levels. It is also assumed that μ is chosen so that $\sum \tau_i = 0$, $i = 1, \dots, k$ holds. This is the fixed effects model. [26](#)

fractional factorial experiment It is a subset of a [factorial experiment](#).

G

generative artificial intelligence Generative artificial intelligence is a type of [artificial intelligence](#) that can create new content, such as text, images, music, or code. It can learn from existing [data](#) and then generate something new, original, and human like. It uses [generative models](#) to produce text, images, videos, or other forms of [data](#). These models learn the underlying patterns and structures of their [training data set](#) and use them to produce new [data](#) based on the input, which often comes in the form of [natural language](#) prompts. A generative [artificial intelligence](#) system is constructed by applying [unsupervised machine learning](#) invoking [artificial neural network](#) architectures (GAN, VAE, Transformer) or [self supervised machine learning](#) to a [data set](#). Generative artificial intelligence can be [unimodal training](#) or [multimodal training](#). [19](#), [25](#), [26](#)

generative model In [statistical classification](#) the generative approach computes classifiers such as the naive Bayes classifier or the linear discriminative analysis. A discriminative model is a model of the [conditional probability](#) $P(X|Y = y)$ of the observable X given a target Y . A generative model can be used to generate random instances (outcomes) of an [observation](#) x . [23](#)

GitHub It is a cloud-based platform where you can store, share, and work together with others to write code.

GitLab It is a cloud-based platform where you can store, share, and work together with others to write code. 24

graphical user interface A graphical user interface (GUI) is a visual interface that allows users to interact with a computer or other electronic device using icons, buttons, and menus. GUIs are interactive and replace text-based commands with user-friendly actions. The goal of a GUI is to make decision points easy for users to find, understand, and use.

H

hand geometry TBD

heartbeat A heartbeat is one complete pulsation of the heart.

Heptapod FOSS It is a web-based platform designed to bring Mercurial support to [GitLab](#).

hidden layer It is a synthetic layer in a [artificial neural network](#) between the input layer (the [exogenous variables](#)) and the output layer (the [prediction](#)). Hidden layers typically contain an activation function (such as ReLU) for training. An [artificial neural network](#) contains more than one hidden layer. 21

hubris It describes a personality quality of extreme or excessive pride or dangerous overconfidence, often in combination with (or synonymous with) arrogance. Hubris often indicates a loss of contact with reality and an overestimation of one's own competence, accomplishments or capabilities.

human language It is a structured system of communication that consists of grammar and vocabulary. It is the primary means by which humans convey meaning, both in spoken and signed forms, and may also be conveyed through writing. Human language is characterized by its cultural and historical diversity, with significant variations observed between cultures and across time. Human languages possess the properties of productivity and displacement, which enable the creation of an infinite number of sentences, and the ability to refer to objects, events, and ideas that are not immediately present in the discourse. The use of human language relies on social convention and is acquired through learning. 26

hypothesis test It is a method for proving that an observed difference was not due to random chance.

I

identity and access management Identity and access management (IAM) is a framework of policies and technologies to ensure that the right users have the appropriate access to technology resources. It identifies, authenticates, and controls access for individuals who will be using information technology resources, hardware, and access to applications. 27

ImageNet The ImageNet project is a large visual database designed for use in visual object recognition software research. More than 14 million images have been hand-annotated by the project to indicate what objects are pictured and in at least one million of the images, bounding boxes are also provided. ImageNet contains more than 20,000 categories, with a typical category, such as "balloon" or "strawberry", consisting of several hundred images. The database of annotations of third-party image URLs is freely available directly from ImageNet, though the actual images are not owned by ImageNet. Since 2010, the ImageNet project runs an annual software contest, the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), where software programs compete to correctly classify and detect objects and scenes. The challenge uses a "trimmed" list of one thousand non-overlapping classes.

individuals chart (X) It is a [process behaviour chart](#) that plots the single values taken in a rational manner. The chart also has an upper [control chart limit](#) line, a lower [control chart limit](#) line, and an [average](#) line. 20

ineffable It means incapable of being expressed in words.

instance It is an example, an [observation](#), or a row of a spreadsheet.

interaction It is a dependent relation between two factors. In a [designed experiment](#), if the [effects](#) of factor A are not the same at all levels of factor B, then A and B interact.

interest TBD 22

interquartile range The interquartile range (IQR) is the value of the 75th percentile minus the value of the 25th percentile. This [dispersion statistic](#) attempts to measure the variability of points near the centre. 22

intraclass correlation coefficient The intraclass correlation coefficient (ICC) describes how strongly units in the same group resemble each other, with and without operators.

iris The iris is the opaque contractile diaphragm perforated by the pupil and forming the colored portion of the eye.

J

JAX It is a Python library for accelerator oriented array computation and program transformation, designed for high performance numerical computing and large-scale [machine learning](#). It was developed by Google. 25

joint probability distribution TBD

K

K-fold cross-validation It is a [machine learning](#) technique that divides all the samples in groups of samples, called folds (if , this is equivalent to the Leave One Out strategy), of equal sizes (if possible). The prediction function is learned using folds, and the fold left out is used for test. Step 1. Split the dataset into K equal partitions or folds. Step 2. Use fold 1 as the testing set and the union of the other folds as the training set. Step 3. Calculate testing accuracy. Step 4. Repeat steps 2-3 K times, using a different fold as the testing set each time. Step 5. Use the average testing accuracy as the estimate of out-of-sample accuracy.

K-means clustering It is an algorithm that divides [n observations](#) into k clusters, in which each [observation](#) belongs to the cluster with the nearest mean. This algorithm clusters [data](#) by trying to separate [observations](#) into k clusters of equal variance, minimizing the within-cluster sum-of-squares.

k-nearest neighbour The k-nearest neighbour (k-NN) is a non-parametric supervised learning method used in classification and prediction. The input consists of the k closest training examples in a [data set](#). The output depends on whether k-NN is used for classification or prediction.

Keras It is [open-source software](#) for [deep learning](#) with an [application programming interface](#) that makes it very simple to train and run [artificial neural networks](#). It can run on top of [TensorFlow](#) and other [software frameworks](#) (JAX, PyTorch). [31](#)

keyword argument Keyword arguments (named arguments) are values that, when passed into a function, are identifiable by specific [parameter](#) names. A keyword argument is preceded by a [parameter](#) and the assignment operator, =.

L

label It is an answer, a target, a ground truth, or an output.

labeled data Labeled [data](#) is a collection of raw [data](#) that has been assigned labels to provide context and meaning. Labels can be used to categorize [data](#), such as by indicating whether an image contains a cat or dog, or if an audio recording includes a specific word. [30](#)

large language model A large language model (LLM) is a type of artificial intelligence model that is trained on a massive amount of text [data](#). It is capable of generating human quality text, translating languages, writing different kinds of creative content, and answering one's questions in an informative way. Thus it is a form of [generative artificial intelligence](#). It is a specific type of [artificial intelligence](#) model within the [natural language processing](#) field. A large language model requires very large [data sets](#). [31](#)

LaTeX LaTeX, pronounced "lay-tech," is a high-quality typesetting system; it includes features designed for the production of technical and scientific documentation. LaTeX is the de facto standard for the communication and publication of scientific documents. LaTeX is available as free software (Oetiker et al. [2023](#)).

linear regression It is a linear approach for modeling the relationship between a scalar [endogenous variable](#) and [exogenous variable](#). The case of one [exogenous variable](#) is called simple linear regression. The case of more than one [exogenous variable](#) is called multiple linear regression.

location It is the central value of a sample of values or distribution. [21](#), [22](#)

location statistic It is a statistic that estimates the central value that best describes the [data](#). Examples include the [average](#), [median](#), and [mode](#). [20](#)

logistic regression In [statistic](#), the (binary) logistic model (or logit model) is a statistical model that models the probability of one event (out of two alternatives) taking place by having the log-odds (the logarithm of the odds) for the event be a linear combination of one or more independent variables (predictors). In regression analysis, logistic regression (or logit regression) is estimating the parameters of a logistic model (the coefficients in the linear combination).

M

machine learning It is a field of study that gives computers the ability to learn without being explicitly programmed ((Samuel [1959](#))). It is an approach to making repeated decisions that involves algorithmically finding patterns in [data](#) and using these to make models that deal correctly with new [data](#). There are four types of machine learning: [supervised machine learning](#), [unsupervised machine learning](#), [semi-supervised learning](#), and [reinforcement learning](#). A common task in machine learning is to build a model from [data sets](#). There are three [data sets](#) commonly used in the different stages of the creation of the model: [training data set](#), [validation data set](#), and [test data set](#). [6](#), [19](#), [21–26](#), [29–32](#)

machine learning engineer This individual understands how to use algorithms to create models. Their expertise at wrangling code gets existing algorithms to accept and churn through [data](#) to create models from old [data](#) to succeed with new [data](#). This individual can code quickly and can cope with failure. You cannot

know in advance what what will work, so you cannot expect to get a perfect result on the first go. This individual iterates towards a solution. 21

machine learning hyperparameter In [machine learning](#) it is a [machine learning parameter](#) that can be set in order to define any configurable part of a model's learning process. Hyperparameters can be classified as either model hyperparameters or algorithm hyperparameters. 31, 32

machine learning parameter In [machine learning](#) it is a characteristic that the model learns from the [data set](#). 26

machine translation TBD

main effect It is the effect of a single factor in a [designed experiment](#).

Matplotlib Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. It is widely used for [data](#) visualization and plotting in the [Python programming language](#).

mean absolute error The mean absolute error (MAE) is the mean of the absolute value of the errors.

$$\text{MAE} = \frac{\sum_{i=1}^n (|y_i - \hat{y}_i|)}{n}$$

mean squared error The mean squared error (MSE) is the mean of the squared errors.

$$\text{MSE} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

Measurement System Analysis Measure system analysis (MSA) is a method to evaluate measurements systems, to understand the causes of measurement variation

median It is the value of the point which has half the [data](#) smaller than that point and half the [data](#) larger than that point. There are various formulae to calculate the median, for example JMP, Minitab, and SAS estimate it in different ways. 25

median absolute deviation The median absolute deviation (MAD) is a variation of the [average absolute deviation](#) that is even less affected by extremes in the tail because the [data](#) in the tails have less influence on the calculation of the median than they do on the [average](#).

$$\text{MAD} = \text{median}(|Y_i - \tilde{Y}|)$$

22

metaphor A metaphor is the use of a word or phrase to refer to something other than its literal meaning, invoking an implicit similarity between the thing described and what is denoted by the word or phrase. 31

mixed effects model It is a statistical model contain both [fixed effects model](#) and [random effects model](#).

mode It is the value of the random sample that occurs with the greatest frequency. 25

moving range chart (mR) It is a [process behaviour chart](#) that plots the difference between successive pairs of values taken in a rational manner. The chart also has an upper [control chart limit](#) line, a lower [control chart limit](#) line, and an [average](#) line. 32

multimodaltraining Multimodal training refers to the process of teaching a [machine learning](#) model to process and understand [data](#) from multiple sources, like text, images, and audio (modalities), allowing it to learn relationships between these various [data](#) types and create a comprehensive understanding of the information presented; essentially, training a model to analyze and interpret the world through multiple senses, similar to how humans do. 23, 32

N

naive Bayes classifier It is a family of simple probabilistic classifiers based on applying Bayes' theorem with strong (naive) independence assumptions between the features.

natural language A natural language is any language that occurs naturally in a human community by a process of use, repetition, and change without conscious planning or premeditation. 19–21, 23, 26, 28

natural language processing [Natural language](#) processing is a field of artificial intelligence that focuses on the interaction between computers and [human language](#) ([natural language](#)). It involves writing [computer programs](#) to understand, interpret, and generate [human language](#). It is a broad field that encompasses various techniques and applications related to [natural language](#). It is a form of [generative artificial intelligence](#). Some examples of natural language processing models are [sentiment analysis](#), [text classification](#), and [topical model](#). 19, 25, 29, 30

Nelson's Rules They are a set of [process behaviour chart](#) rules for assessing statistical control.

normal distribution It is a type of continuous probability distribution for a real-valued random variable

O

observation It is the careful, close examination of an object, process, or other phenomenon for the purpose of collecting data about it or drawing conclusions. 20, 22–25, 27, 29, 30

open-source software It is computer software that is released under a license in which the copyright holder grants users the rights to use, study, change, and distribute the software and its source code to anyone and for any purpose. Open-source software may be developed in a collaborative public manner. Open-source software is a prominent example of open collaboration, meaning any capable user is able to participate online in development, making the number of possible contributors indefinite. The ability to examine the code facilitates public trust in the software. 19, 25, 29–31

openpyxl It is a **Python** library to read/write Excel 2010 xlsx/xlsm/xltx/xltm files. ((Gazoni and Clark 2024))

operating system An operating system is software that manages computer hardware and software resources, and provides common services for **computer programs**. 22

outlier It is a **datum** point that differs significantly from other **observations**. An outlier may be due to variability in the measurement or it may indicate experimental error. An outlier can cause serious problems in statistical analyses.

overfitting It is the production of an analysis that corresponds too closely or exactly to a particular set of **data**, and may therefore fail to fit to additional **data** or predict future **observations** reliably. An overfitted model is a mathematical model that contains more parameters than can be justified by the **data**. The essence of overfitting is to have unknowingly extracted some of the residual variation (that is, the noise) as if that variation represented underlying model structure. The possibility of over-fitting exists because the criterion used for selecting the model is not the same as the criterion used to judge the suitability of a model. For example, a model might be selected by maximizing its performance on some set of **training data set**, and yet its suitability might be determined by its ability to perform well on unseen data; then over-fitting occurs when a model begins to “memorize” **training data set** rather than “learning” to generalize from a trend. As an extreme example, if the number of parameters is the same as or greater than the number of **observations**, then a model can perfectly predict the **training data set** simply by memorizing the **data** in its entirety. Such a model, though, will typically fail severely when making predictions.

P

p-hacking It is the use of **descriptive analytics** to uncover patterns in **data** that can be presented as statistically significant, without first devising a specific hypothesis about the underlying causality.

package manager A package manager is software that handles package installation, removal, and downgrades of software for the host system. (**Package manager**) 13, 27

pacman Pacman is an **Arch Linux package manager** that handles package installation, removal, downgrades, and automatic dependency resolution. (**Pacman**) 13

pandas It is an open source, BSD-licensed library providing high-performance, easy-to-use **data** structures and **data** analysis tools for the **Python programming language**. ((pandas.pydata.org 2024))

parallelism chart It is a **process behaviour chart** that plots the lines from the average chart and puts them on top of each other. It is used in measurement system analysis. Differences for any part indicate an operator bias relative to other operators.

parameter Parameters are the input variables bounded by parentheses when defining a Python function. 19, 25, 27

polymath A polymath is an individual whose knowledge spans many different subjects, known to draw on complex bodies of knowledge to solve specific problems. 19

positional argument Positional arguments are values that are passed into a function based on the order in which the **parameters** were listed during the function definition.

Power BI It is a collection of software services, apps, and connectors designed to transform **data** into clear and actionable insights. It falls under the umbrella of Business Intelligence (BI) tools, which help businesses analyze **data** to make informed decisions. ((Microsoft 2024))

PowerShell PowerShell is a task automation and configuration management framework from Microsoft, consisting of a command-line shell and a scripting language. 14

prediction It is the output of a model. 24

privileged access management Privileged access management (PAM) is a type of **identity and access management** and **cybersecurity** technology that focuses on the control, monitoring, and protection of privileged accounts in an organization. Accounts with privileged access grant users enhanced permissions, making them prime targets for attackers due to their extensive access to vital systems and sensitive **data**.

probability distribution In **probability theory** and **statistics**, a probability distribution is the mathematical function that gives the probabilities of occurrence of possible outcomes for an **experiment**. It is a mathematical

description of a random phenomenon in terms of its **sample space** and the probabilities of event (subsets of the **sample space**). 31

probability theory TBD 27

probable error It is also called resolution. It is the interval within which a single measurement might occur 50 % of the time, $\pm 0.6745 \times \sigma_{\text{pure error}}$.

process behaviour chart TBD 20, 24, 26, 27, 29, see also **control chart**

process capability It is the ability of a process to meet a performance standard.

process capability index It is the ratio of a specification measure and a **dispersion** measure.

process capability index loss function (random sample) The process capability index loss function (random sample) (C_{pm}) calculates process capability with respect to deviation from the **average** and deviation from the target.

process capability index one-sided (random sample) The process capability index one-sided (random sample) (P_{pk}) calculates the distance from each specification limit to the **average**, compares them to the width of the process variation, and reports the smaller value.

process capability index one-sided (rational sample) The process capability index one-sided (rational sample) (C_{pk}) calculates the distance from each specification limit to the **average**, compares them to the width of the process variation, and reports the smaller value.

process capability index two-sided (random sample) process capability index two-sided (random sample) (P_p) compares the width of the process specification to the width of the process variation

process capability index two-sided (rational sample) The process capability index two-sided (rational sample) (C_p) compares the width of the process specification to the width of the process variation.

programming language A programming language is a system of notation for writing **computer programs**. 20, 26–28, 30

prompt engineering It is the practice of carefully crafting and optimizing the input text in a **natural language**, called a *prompt*, given to a generative **artificial intelligence** model to ensure it produces the most accurate, relevant, and desired output.

PyAutoGUI It lets your **Python scripts** control the mouse and keyboard to automate interactions with other applications. The API is designed to be simple. It works on Windows, macOS, and Linux, and runs on **Python 2** and 3." ((Sweigart 2023))

Python Python is an easy to learn, powerful **programming language**. It has efficient high-level **data** structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms. (Python Software Foundation 2024) 17, 26–28

Python attribute A Python attribute is a variable associated with an object. It represents the **data** or state of an object. 28

Python class A Python class is a blueprint for creating objects. It defines the properties and methods (structure) that will be shared by all instances of that class. 28

Python distribution A Python distribution is a software bundle containing a Python interpreter and the Python standard library. It can be installed from the **Python.org distribution page** or using the package manager of various operating systems.

Python function A Python function is a reusable block of code that performs a specific task. It takes input, processes it, and returns an output. 28

Python library A **Python** library is a collection of code modules to extend the functionality of the **Python programming language**. These modules contain functions, classes, and variables that can be used to perform specific tasks.

Python method A Python method is a **Python function** defined with a **Python class**. It operates on the **data** (**Python attribute**) associated with an objects of the class.

Python namespace It is the place where a variable is stored. Namespaces are implemented as dictionaries. There are local, global, and built-in namespaces as well as nested namespaces in objects (in methods). Namespaces support modularity by preventing naming conflicts. For instance, the functions `builtins.open` and `os.open()` are distinguished by their namespaces. Namespaces also aid readability and maintainability by making it clear which module implements a function. For instance, writing `random.seed()` or `itertools.islice()` makes it clear that those functions are implemented by the `random` and `itertools` modules, respectively.

Python script A **Python** script is a plain text file of **Python** commands that is executed on a compute terminal, also known as a shell. A **Python** script is typically saved in a plain text file with a `.py` extension. To run a **Python** script, you need a **Python** interpreter installed on your computer. The interpreter reads the code in the script line by line, translates it into machine code the computer understands, and executes the instructions. 28

Python Tesseract Python Tesseract is a wrapper for **Tesseract**. It is also useful as a stand-alone invocation script

to tesseract, as it can read all image types supported by the Pillow and Leptonica imaging libraries, including jpeg, png, gif, bmp, tiff, and others. Additionally, if used as a script, Python Tesseract will print the recognized text instead of writing it to a file. ((Lee 2024))

PyTorch It is an [open-source software framework](#) for [machine learning](#) and [artificial intelligence](#). It can be used across a range of tasks but has a particular focus on [natural language processing](#) and [computer vision](#). It was developed by Meta. 25

R

random effects model It is also called a variance components model. It is a statistical model where the model parameters are random quantities. The mathematical model that describes the relationship between the response and treatment for the one-way ANOVA is given by $Y_{ij} = \mu + \tau_i + \epsilon_{ij}$, where Y_{ij} represents the j^{th} [observation](#) ($j = 1, 2, \dots, n_i$) on the i^{th} treatment ($i = 1, 2, \dots, k$ levels). μ is the common effect for the whole [experiment](#), τ_i represents the i^{th} treatment effect, and ϵ_{ij} represents the random error present in the j^{th} [observation](#) on the i^{th} treatment. The errors ϵ_{ij} are assumed to be normally and independently (NID) distributed, with mean zero and variance σ_e^2 . μ is always a fixed parameter, and $\tau_1, \tau_2, \dots, \tau_k$ are considered to be random parameters if the levels of the treatment are random and not a fixed sample from a population of possible levels. It is also assumed that μ is chosen so that $\sum \tau_i = 0$, $i = 1, \dots, k$ holds. This is the random effects model. 26

random forest It is an ensemble learning method for classification, prediction, and other tasks that operates by constructing a multitude of decision trees.

randomisation It is a reordering of treatments in a [designed experiment](#) from the order shown in an orthogonal array in order to avoid the effect of unknown factors and the bias due to sequence.

range It is the difference between the largest and smallest value in a sample. $X_{\max} - X_{\min}$ 20, 22, 29

range chart (R) It is a [process behaviour chart](#) that plots the [range](#) of a subgroup of values taken in a rational manner. The chart also has an upper [control chart limit](#) line, a lower [control chart limit](#) line, and an [average](#) line. 32

reinforcement learning It is an area of [machine learning](#) concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward. 25

Requests It is a simple, yet elegant, HTTP library. Requests allows you to send HTTP/1.1 requests extremely easily. ((pypi.org 2023))

residual error It is the difference between the actual response value obtained in a treatment of a [designed experiment](#) and the estimate value from a regression model of factors ((Box, W. G. Hunter, and J. S. Hunter 1978, pp. 333–334)).

resolution It is a method to categorize the amount of confounding in an [experiment](#). The higher the resolution number, the less the confounding. The resolution of a two-level fractional factorial design is the length of the shortest word in the defining relation ((Box, W. G. Hunter, and J. S. Hunter 1978, pp. 385–390)).

resolution III design Main effects are foundounded with two-factor interactions ((Box, W. G. Hunter, and J. S. Hunter 1978, pp. 385–390)).

response graph It is a plot of the response versus the levels for a factor.

retina The retina is the sensory membrane that lines the eye, is composed of several layers including one containing the rods and cones, and functions as the immediate instrument of vision by receiving the image formed by the lens and converting it into chemical and nervous signals which reach the brain by way of the optic nerve.

robotic process automation Robotic process automation (RPA) uses software robots to handle repetitive computer tasks. The benefits include reduced execution time, customization for specific needs, perform complex workflows, and consistent repeatability to eliminate human errors.

root-mean-squared error The root-mean-squared error (RMSE) is a frequently-used measure of the differences between values predicted by a model or an estimator and the values observed.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y})^2}{n}}$$

S

sample space It is the set of all possible outcomes of a random phenomenon being observed. 28

science Science is a systematic discipline that builds and organizes knowledge in the form of testable hypotheses and predictions about the universe. It establishes objective truths, not by wishful or magical thinking, but by repeated and verified experiments. This makes such truths true, whether or not you believe in them. 1

scientific method It is an [empirical method](#) consisting of a set of procedures, guidelines, assumptions, and attitudes required for the organized and systematic collection, interpretation, and verification of data and the discovery of reproducible evidence, enabling laws and principles to be stated or modified. One researcher comes up with a result (from theory, hunch, [observation](#)). This is not the truth. A scientific, emergent truth is not the result of any one [experiment](#). Many researchers try to verify it. If no one can prove it wrong, one can declare a consensus of [observations](#) and [experiments](#) has emerged in the scientific community. [9](#)

scikit-learn It is an [open-source software](#), package for [machine learning](#) that supports supervised and unsupervised learning. It also provides various tools for model fitting, [data](#) preprocessing, model selection, model evaluation, and many other utilities.

Scrapy Scrapy is a fast high-level web crawling and [web scraping](#) framework, used to crawl websites and extract structured [data](#) from their pages. It can be used for a wide range of purposes, from [data](#) mining to monitoring and automated testing. ((scrapy.org [2024](#)))

seborrheic keratosis It occurs when skin cells, keratinocytes, multiply rapidly, resulting in a non-cancerous growth. This slow-growing spot is typically raised and may have a rough texture, and can appear anywhere on your body. It is neither viral or bacterial and cannot spread. It is most common in adults over 50 years of age. See [Yale Medicine—Seborrheic Karatosis](#). [6](#)

self supervised machine learning Self supervised machine learning is a [machine learning](#) technique that trains models to create their own labels from [unlabeled data](#) data. This method is also known as predictive or pretext learning. It is useful because it allows models to be trained using large amounts of [unlabeled data](#) data, which is often cheaper and more abundant than [labeled data](#). This makes SSL a cost-effective and time-efficient way to train models, especially in fields like computer vision and [natural language processing](#) where [labeled data](#) can be scarce. Self supervised learning creates its own “supervisory signals” by manipulating the [data](#) itself to generate learning tasks, effectively acting like a supervised learning approach within an unsupervised framework. [23](#)

semi-supervised learning It is an approach to [machine learning](#) that combines a small amount of [labeled data](#) to guide the model with a large amount of [unlabeled data](#) data during training. Semi-supervised learning falls between unsupervised learning (with no labeled [training data set](#)) and supervised learning (with only labeled [training data set](#)). It is a special instance of weak supervision. [25](#)

sentiment analysis TBD [26](#)

shell This is a program that interprets the commands typed into a [terminal](#). [31](#)

shell script A shell script is a plain text file of commands that is executed on a compute terminal, also known as a shell. Linux (gnu.org [2024](#)), citezsh, Mac (Stephenson [2024](#)), and Windows PowerShell (**spowershell**) can easily execute shell scripts. Windows PowerShell offers similar functionality and has its own syntax for the commands.

Shewhart control chart TBD see also [control chart](#)

sicario A sicario is a hired gunman or assassin, especially in Latin America.

software Software consists of [computer programs](#) that instruct the execution of a computer. It includes design documents, specifications, and documentation.

software framework It is a collection of code for building applications. It saves time and effort by providing reusable components and functionalities. It typically consists of classes and functions, interfaces and protocols, and configuration options. [25](#), [29](#), [31](#)

software robot It is a software program that automates repetitive, rule-based tasks on a computer.

source code Source code is a plain text [computer program](#) written in a [programming language](#). A programmer writes the human readable source code to control the behavior of a computer.

sources of variation These are the factors in a process that cause variation on the dependent variable. These factors may vary in a deliberate manner (fixed effects), random manner (random effects), or mixed effects (fixed and random factors)

special cause Variation due to special cause is unexpected and unusual variation. The cause can be identified. [21](#)

specification In the context of measurement system analysis, process capability, and process control, a specification is a lower limit, a target, and an upper limit.

standard deviation It is the square root of the variance.

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

[20-22](#)

standup paddle board It is a large, surfboard-like vessel that allows a person to stand upright and use a single-bladed paddle to propel themselves across the water. Standup paddle boards come in various types, including solid and inflatable models, and are used for a wide range of activities such as flatwater paddling, standup paddle board surfing, paddleboard yoga, fishing, and racing on lakes, rivers, and oceans.

statistic TBD 21–23, 25

statistical classification TBD 22, 23

statistical control A process is said to be in a state of statistical control if the variation for both charts is stable and predictable. This means that no points fail various control chart rules, such as Nelson's rules, and no systematic patterns of variation are present. It is also called process stability. 21

statistical inference It is the process of using **data** analysis to infer properties of an underlying distribution of probability. Inferential statistical analysis infers properties of a population, for example by testing hypotheses and deriving estimates. It is assumed that the observed **data set** is sampled from a larger population.

statistician This individual comes to conclusions safely beyond the **data** being analyzed. It involves a rigour and a careful approach, to prevent making unwarranted conclusions. 21

statistics TBD 27

stochastic parrot In **machine learning**, term stochastic parrot is a **metaphor** to describe the theory that **large language models**, though able to generate plausible language, do not understand the meaning of the language they process.

supervised machine learning It is the **machine learning** task of learning a function that maps an input to an output based on example input–output pairs. The five steps of supervised learning are: 1. import the classes you plan to use, 2. instantiate the transformers, classifiers, regressors, and selectors, 3. train the model on an existing **data set**, 4. tune the **machine learning hyperparameters**, and 5. predict the response for a new **data set**. 25

support-vector machine A support-vector machine (SVM) is a supervised learning model with associated algorithms that analyze **data** for classification and prediction.

Systems, Applications, and Products Systems, Applications, and Products (SAP) is a suite of software solutions developed by the German company SAP SE. SAP software is primarily focused on Enterprise Resource Planning (ERP), which means it helps businesses manage and integrate various core functions.

T

Taguchi Loss Function It is a graphical depiction of loss to describe a phenomenon affecting the value of products produced by a company. It was developed by the Japanese business statistician Genichi Taguchi.

tautology It is a formula or assertion that is true in every possible interpretation.

$$X = Y \text{ or } X \neq Y$$

is true in every possible interpretation. "Either the ball is green or the ball is not green" is always true in every possible interpretation.

taxes TBD 22

TensorFlow It is an **open-source software software framework** for **machine learning** and **artificial intelligence**. It can be used across a range of tasks but has a particular focus on training and inference of **artificial neural networks**. It was developed by Google Brain, now Google DeepMind. It provides an **application programming interface** called **Keras**. It is also described as a library for computation using data flow graphs for scalable **machine learning**. 25

terminal It is the interface that interacts with the **shell**. It is the windows where one types commands and receives the output. 30

Tesseract It is an optical character recognition engine (libtesseract) and a command line program (tesseract). ((Tesseract OCR 2024)) 28

test data set It is a **data set** that is independent of the **training data set** but follows the same **probability distribution** as the **training data set**. It is used to assess the performance of the **endogenous variable**. 21, 25

test-retest error It is the consistency of operators making repeated measurements of the same characteristic on the same part with the same measuring instrument. It is a measure of the common cause variation inherent in the measurement system. It also is used to describe the consistency of a single instrument in making repeated measurements of the same part.

text classification TBD 26

text summarization TBD

topical model TBD 26

training data set It is a **data set** of examples used during the learning process and is used to fit the **exogenous variables** to the **endogenous variables**. 23, 25, 27, 30, 31

treatment It is a specific combination of factors with one level from each factor. It is also called a run.

type I error It is incorrectly rejecting a null hypothesis. It is the probability of saying something is different when in fact it is not. We find a difference that isn't really there. It is also called alpha risk, producer's risk, and false positive.

type II error It is incorrectly failing to reject a null hypothesis. It is the probability of saying something is not different when in fact it is. We fail to find a difference when there is one. It is also called beta risk, consumer's risk, and false negative. It is used to calculate the power of the test (1-beta), as well as in calculations of sample size.

type III error It is correctly rejecting the wrong null hypothesis.

U

underfitting It occurs when a mathematical model cannot adequately capture the underlying structure of the **data**. An under-fitted model is a model where some parameters or terms that would appear in a correctly specified model are missing. Under-fitting would occur, for example, when fitting a linear model to non-linear **data**. Such a model will tend to have poor predictive performance.

unimodal training Unimodal training refers to the process of training a **machine learning** model using only one type of **data** (modality), such as text, images, or audio, as opposed to **multimodal training** which combines multiple **data** types simultaneously; essentially, a unimodal model focuses on learning patterns from a single **data** source to make predictions. 23

unlabeled data Unlabeled **data** is raw **data** that doesn't have any explicit tags or categories. It's a collection of samples of items, such as images, videos, articles, or social media posts, that are not labeled or explained. 30, 32

unsupervised machine learning Unsupervised machine learning is a type of **machine learning** that uses algorithms to analyze **unlabeled data** and discover patterns without human intervention. In unsupervised learning, a model is given input **data** and is allowed to identify patterns and relationships on its own without any explicit labels or goals. 23, 25

urllib It is a package that collects several modules for working with URLs. ((pypi.org 2024))

use case It is a usage scenario for a piece of software. It suggests where a piece of software may be useful.

V

validation data set It is a **data set** of examples used during the learning process to tune the **machine learning hyperparameters** of a model. 25

variance It is a measure of **dispersion**, how far a set of numbers is spread out from their **average**. It is the expectation of the squared deviation of a random variable from its population **average** or sample **average**.

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

20

virtual assistant TBD

virtual human It is a computer generated human like figure that can be used for various purposes such as: customer service, virtual assistance, or entertainment. 22

voiceprint A voiceprint is an individually distinctive pattern of certain voice characteristics that is spectrographically produced.

W

web scraping It is the practice of gathering **data** through any means other than a program interacting with an API (or, obviously, through a human using a web browser). This is most commonly accomplished by writing an automated program that queries a web server, requests **data** (usually in the form of HTML and other files that compose web pages), and then parses that **data** to extract needed information. ((Mitchell 2024)) 30

Windows Task Scheduler Windows Task Scheduler is a built-in utility that allows you to automate the execution of programs, scripts, or various tasks at specific times or intervals on a Windows computer.

within-operator error It is the variation of measurements made by one operator on all parts

within-part error It is the variation of measurements within a sample for a single part, over time. Several measurements are taken in different locations on a part and this is repeated at specific time intervals.

within-subgroup error It is the variation within a sample of parts. If the sample is carefully chosen, this variation is the background error of a process. It is shown on the **moving range chart (mR)** or the **range chart (R)**. The **average range** is calculated to estimate the overall error.

woke TBD

Endnotes

Notes

Notes for Chapter 1

1. See Smith (1982) for an architectural breakdown of early mainframe limitations.

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Index

automation, [6](#)

engineering, [9](#)

memory constraints, [6](#)

software, [9](#)

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Gilles Pilon, P.Eng., is a Senior Machine Learning Engineer and Data Scientist with an extensive background in engineering frameworks and advanced systems development. A graduate of McMaster University in Metallurgical Engineering, he specializes in high-performance computing architectures, custom automation pipelines, and robust software infrastructure.

When not structuring complex codebases or optimizing technical documentation frameworks, he is an avid cyclist focused on performance analytics and outdoor endurance. He lives with his family, continuing to build open-source tools and data utilities.

